

Cloudscape Design System Analysis

For AWS APM Claude Plugin | Hybrid Renderer Visualization Layer

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Plugin: aws-apm v0.2.1

Source: cloudscape.design + Custom Color Library Screenshots

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1. Executive Summary

This analysis examines the Cloudscape Design System (AWS's open-source design system powering 220+ AWS products) and maps it against the custom color library you've built for the AWS APM Claude plugin's hybrid renderer. The goal is to identify which Cloudscape components, patterns, and tokens can be leveraged to build richer, more consistent APM visualizations through the plugin's manifest-driven rendering pipeline.

Key finding: Your color library is already well-aligned with Cloudscape's dark-mode palette. The 6 seed color scales (Neutral, Primary, Error, Success, Info, Amber) map cleanly to Cloudscape's semantic token categories. The 16 data visualization series colors provide chart coverage that Cloudscape itself handles via a similar numbered series system. The renderer's existing 7 widget types cover the core APM use cases, but there are clear opportunities to add widgets inspired by Cloudscape's chart components (area, bar, line, pie/donut) and status patterns (status indicator, progress bar, flashbar).

2. Your Color Library Analysis

Extracted from the 5 screenshots of your Design System Color Library (Dark Mode):

2.1 Seed Colors

Role	Hex	Usage
Primary	#7B2CF5 (purple)	Interactive elements, focus rings, toggles, checkboxes
Error	#FF375D (pink-red)	Error states, blocked/degraded, outage indicators
Success	#1FCE80 (green)	Active/operational, healthy status, success states
Info	#2196F7 (blue)	Informational states, links, focus indicators

2.2 Base Color Scales (100-900)

Six 9-step ramps: Neutral (#F3F3F3 to #0F141A), Primary (#E8C0FF to #2F0066), Error (#FFD5D5 to #4A0012), Success (#C0FFD5 to #003D1A), Info (#C0E0FF to #002B66), Amber (#FFE0A0 to #6B3D00). Each ramp provides fine-grained contrast control for backgrounds, borders, text, and interactive states at every elevation level.

2.3 Semantic Tokens

Category	Tokens Defined	Notes
Backgrounds & Surfaces	Page bg, Surface 1, Surface 2, Input bg	Dark base: #0F1A1A, surfaces layer up
Text	Primary, Secondary, Muted, Disabled, Info, Error, Success	6 text hierarchy levels
Borders	Border, Border muted, Input border	3 border intensity levels
Primary Action	Single primary action color	Maps to primary-500
Status	Active, Pending, Blocked, Inactive, Error, Draft	6 status states with semantic colors
Buttons	Primary default/hover/active/disabled	Full state coverage for both variants
Toggles & Forms	Toggle on/off, knob, checkbox states	Interactive control states
Focus & Selection	Focus ring, Selected row bg, Link hover	Accessibility-critical tokens
Semantic Text Colors	On primary, Error, Success, Warning, Info	Status-aware text coloring
Status Extended	Active/Pending/Outage/Inactive subtle bg	Subtle background variants for status

2.4 Data Visualization (16 Series Colors)

Series 1-16 provide a carefully chosen palette for multi-series charts. Key pairs: Series 1 (info-blue #4A90D9), Series 2 (primary-purple #8B5CF6), Series 3 (success-green #10B981), Series 4 (amber #F59E0B), Series 5 (error-red #EF4444), Series 6 (cyan #06B6D4). Plus threshold line (white dashed), grid lines (neutral-700). This aligns with Cloudscape's own 12-contrast-level chart color system.

2.5 Gradients & Application Examples

Three gradients defined: Primary gradient (purple to blue), Accent gradient (blue to green), Subtle gradient (neutral-100 to white). Applied as progress bars, banners/alert boxes (3 severity levels), and hero sections. The gradient banners are particularly relevant for the renderer's meta-bar severity strip.

3. Cloudscape Design System Overview

Cloudscape is AWS's open-source design system, created in 2016 and now powering 220+ AWS products. It provides a comprehensive set of React components, design patterns, and foundations for building cloud management interfaces.

Dimension	Count	Details
Components	107	React components, all tested, responsive, accessible
Patterns	74	Reusable component combinations for common UX problems
Demos	33	Full page layout templates showing real implementations
Foundation Topics	13	Colors, typography, spacing, motion, tokens, theming, etc.
Design Tokens	344+	Color tokens alone; plus typography, spacing, motion, borders
Visual Modes	2	Light mode and dark mode with full token coverage
Density Modes	2	Comfortable and compact content density

4. Component Inventory (107 Components)

Grouped by relevance to APM visualization:

4.1 High Relevance for APM Plugin

Component	Why It Matters for APM
Status indicator	Resource health states (healthy/warning/error/critical)
Alert / Flashbar	Severity notifications, alarm transitions, incident banners
Table	Top failing operations, SLO rankings, dependency health
Line chart	Time-series metrics (latency, error rate, throughput)
Area chart	Stacked metric comparisons over time
Bar chart	Service comparisons, error distributions
Pie/donut chart	Budget allocation, error type breakdown
Mixed line+bar chart	Correlating metrics (e.g., latency vs throughput)
Cards / Board	Service health tiles, KPI stat cards
Progress bar	SLO budget burn, error budget remaining
Badge	Severity labels, status tags
Tabs	Multi-view investigation panels
Split panel	Detail drill-down alongside collection
App layout / toolbar	Full APM dashboard structure
Key-value pairs	Trace span attributes, resource properties
Container / Header	Widget grouping with action buttons
Expandable section	Overflow drawer, log detail expansion
Loading bar / Spinner	Async data loading states
Popover / Tooltip	Metric hover details, sparkline values
Steps	Incident timeline progression
Icon	Status icons, severity indicators
Box / Grid / Column layout	Widget arrangement, responsive grid
Board + Board items	Configurable dashboard layout with drag-and-drop

4.2 Medium Relevance

Breadcrumb group (navigation context), Side navigation (APM service tree), Top navigation (global APM controls), Pagination (log viewer paging), Date range picker (time window selection), Property filter (metric filtering), Select/Multiselect (service/region selection), Segmented control (view mode toggle), Token/Token group (filter chips for applied filters), Link (console deep links), Modal (confirmation dialogs), Wizard (guided investigation flows), Code editor/Code view (log lines, query display), Collection preferences (column selection), Annotation context (tutorial overlays for onboarding).

4.3 GenAI Components (New)

Cloudscape now includes dedicated GenAI components highly relevant to a Claude plugin: Avatar (AI entity representation), Chat bubble (conversation display), Prompt input (user query input), Support prompt group (suggested queries), and patterns for Generative AI chat, Artifact previews, Follow-up questions, Thinking states, and Ingress (AI feature entry points).

5. Pattern Library (74 Patterns)

Key patterns relevant to APM plugin visualization:

Pattern	APM Application
Service dashboard	Primary APM landing page with health tiles
Configurable dashboard	User-customizable APM board layout
Data visualization	Chart rendering guidelines and best practices
Details page / as hub	Service detail view with related resources
Split view	Service list + metric detail side panel
Table view	Alarm list, SLO table, operation ranking
Card view	Service health card grid
Filtering patterns	Metric/service/time filtering
Error messages	Failed query handling, MCP error display
Loading & refreshing	Real-time data refresh patterns
Empty states	No SLOs configured, no alarms found
Progressive steps	Multi-step investigation flow
Timestamps	Event timeline formatting
Density settings	Compact vs comfortable metric display
Generative AI chat	Claude conversation interface
Artifact previews	Rendered manifest preview in chat
Thinking	AI reasoning state during investigation
User authorized actions	Approval before destructive operations

33 Demos Available

Cloudscape provides 33 working demos including: Service dashboard, Configurable dashboard, Split view, Table view (client + server-side), Table with inline edit, Table with expandable rows, Details page (3 variants), Card view, Generative AI chat, and more. Each is a full page template showing real component integration.

6. Foundation Layer

6.1 Color System

Cloudscape's color system uses the AWS brand palette with semantic role assignment. Key color families: error (red #db0000/#ff5d64 light/dark), info (blue #006ce0/#539fe5), success (green #00802f/#6aaf3f), warning (yellow #ffe347), severity spectrum (dark-red, red, orange, yellow), neutral (full grayscale #ffffff to #0f141a), and primary (blue #006ce0). Purple #7300e5 is reserved for GenAI labels.

6.2 Visual Modes

Full light and dark mode support via design tokens. Every color token has both a light and dark value. The system uses CSS classes (awsui-dark-mode) and respects prefers-color-scheme. Your plugin's dark-theme-first approach aligns perfectly with Cloudscape's dark mode.

6.3 Content Density

Two modes: comfortable (default, more whitespace) and compact (denser, more data visible). Spacing tokens change between modes. Relevant for the renderer's density budget system (single-focus: 6, investigation: 8, dashboard: 10).

6.4 Typography

Font family: 'Amazon Ember' (internal) / 'Open Sans' fallback. Your renderer already uses 'Open Sans' as primary font. Cloudscape defines heading styles H1-H5, body text, and code text via design tokens. All font sizes, weights, and line heights are tokenized.

6.5 Spacing & Layout

Spacing tokens use an 8px base grid. Key spacing values: xxs(2), xs(4), s(8), m(12), l(16), xl(20), xxl(24). Layout uses Grid component (12-column), Column layout, and container queries. Your renderer already uses container queries (container-name: hr) for responsive behavior.

6.6 Motion

Cloudscape defines motion tokens for transitions: duration-complex (250ms), duration-moderate, easing curves. Motion can be disabled via tokens. Currently not used in your renderer but could enhance widget transitions and hover states.

6.7 Theming

Cloudscape supports theming through design token overrides. Tokens marked 'Themeable' can be customized. This maps directly to your renderer's CSS custom properties approach (--bg-page, --bg-surface, --status-error, etc.).

7. Design Tokens Architecture

Cloudscape uses 344+ color tokens following CTI naming: Category > Type > Item > State. Available as Sass (`$color-text-body-secondary`), JS (`colorTextBodySecondary`), and JSON.

Token Category	Count	Format
Color	344+	{ light: string, dark: string }
Typography (font)	~20	string
Line height	~10	string
Border radius	~8	string
Spacing	~30	{ comfortable: string, compact: string }
Motion	~15	{ default: string, disabled: string }

The JSON format follows the Design Tokens Community Group spec and can be converted to any platform (CSS, Android XML, iOS Swift) via `style-dictionary`. Package: `@cloudscape-design/design-tokens` (npm).

8. Gap Analysis: Current Renderer vs Cloudscape

Aspect	Your Renderer Now	Cloudscape Offers	Gap
Color tokens	14 CSS vars (--bg-page etc.)	344+ semantic tokens	Expand to full token set
Widget types	7 (stat, sparkline, timeline, table, trace, comp, page)	167, 100+ (charts)	Add chart types, status indicator, progress bar
Charts	Sparkline only (SVG points)	Line, area, bar, pie, donut, mixed, scatter, map	Major gap: need real chart widgets
Layout	3 shells (single/investigation/dashboard)	App layout, board, grid, split panel	Add configurable board layout
Status system	5 statuses in stat_card	Full status indicator component with 8+ states	Align to Cloudscape status vocabulary
Dark mode	Dark only	Light + dark via tokens	Add light mode support
Density	Density budget (6/8/10)	Comfortable + compact modes	Consider compact mode toggle
Typography	Open Sans, 3 sizes	Full type scale via tokens	Expand heading hierarchy
Motion	None	Full motion token system	Add transitions for widgets
Accessibility	Basic (semantic HTML)	WCAG AA compliant components	Add ARIA labels, focus management
Interactivity	Sort/search on tables	Full CRUD, drag-drop, inline edit	Add drill-down, board reorder
GenAI patterns	None	Chat, prompts, thinking, artifacts	Add AI investigation flow

9. Mapping Plan: Color Library to Renderer

Your color library's semantic tokens map to the renderer's CSS custom properties. Here's the recommended mapping to bring the renderer in line with your design system:

Your Token	Current Renderer Var	Recommended Update
Page background (#0F1A1A)	--bg-page: #0f1b2a	Update to #0F1A1A
Surface 1 (#192534)	--bg-surface: #192534	Already aligned
Surface 2 (#1F3045)	--bg-surface-2: #1f2d3d	Update to Surface 2 value
Primary text	--text-primary: #e9ebed	Map to your primary text token
Secondary text	--text-secondary: #95a5b8	Map to your secondary text token
Muted text	--text-muted: #6c7c91	Map to your muted text token
Link (#4A90D9)	--link: #539fe5	Update to your link token
Error (#FF375D)	--status-error: #ff5d64	Update to your error seed
Amber (#FFA552)	--status-warning: #ffa552	Already aligned
Success (#1FCE80)	--status-success: #6aaf3f	Update to your success seed
Info (#2196F7)	--status-info: #539fe5	Update to your info seed
Border	--border: #2a3a4f	Map to your border token
Active subtle bg	(not mapped)	Add --status-active-bg
Pending subtle bg	(not mapped)	Add --status-pending-bg
Outage subtle bg	(not mapped)	Add --status-outage-bg
Series 1-16	(not mapped)	Add --chart-series-1 through --chart-series-16
Primary gradient	(not mapped)	Add --gradient-primary for progress bars
Accent gradient	(not mapped)	Add --gradient-accent for success states

10. Recommendations for Plugin Visualization

10.1 Immediate (v0.3.0)

- Update renderer CSS vars to match your color library tokens exactly
- Add chart widgets: line_chart, bar_chart, area_chart (using SVG, no external deps)
- Add status_indicator widget type with Cloudscape-aligned status vocabulary
- Add progress_bar widget for SLO budget burn visualization
- Apply your gradient system to severity banners and progress indicators
- Map all 16 series colors as --chart-series-N CSS vars

10.2 Short-term (v0.4.0)

- Add light mode support using your color scales (100-level backgrounds, 900-level text)
- Implement compact density mode (tighter padding, smaller fonts)
- Add board layout shell for configurable dashboards (drag-and-drop widget reorder)
- Add pie/donut widget for error type and budget breakdowns
- Integrate GenAI patterns: thinking state indicator during investigation
- Add motion tokens for widget entry animations and hover transitions

10.3 Long-term (v0.5.0+)

- Consider adopting @cloudscape-design/design-tokens npm package for full token alignment
- Build split-view investigation pattern: service list + detail panel
- Add inline edit capability for SLO targets and alarm thresholds
- Implement saved filter sets for repeated investigation queries
- Add annotation/tutorial overlay for APM onboarding
- Export design tokens as JSON following the W3C Design Tokens Community Group spec

Sources: cloudscape.design (Components, Foundation, Patterns, Demos, Design Tokens), Custom Color Library screenshots, aws-apm-claude-plugin renderer/styles.css